THE AREA FUNCTION OF THE MERTENS EXCURSION SKELETON (RESEARCH NOTEBOOK, VERSION 7 — MERGED AND AUDITED)

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Abstract. This version merges the complete version 6 notebook with the audited version 7 change set. The principal new results are two elementary lacunarity bounds for long excursions and their combination with the quantitative oscillation theorem of Kaczorowski–Pintz. If $I_\nu = (x_{0,\nu}, x_{1,\nu}]$ denotes a genuine excursion (zero runs between excursions are not themselves excursions), then excursions with $x_{1,\nu} > x_{0,\nu}^A$, $A > 1$, have count $O_A(\log\log X)$ up to starting point X, while excursions with $x_{1,\nu} > \lambda x_{0,\nu}$, $\lambda > 1$, have count at most $(\log\lambda)^{-1}\log X + O(1)$. Kaczorowski–Pintz imply $(\gamma_1/\pi - o(1))\log X$ alternating large values of the Mertens function, of size $x^{\Theta-\varepsilon}$. Consequently RH is equivalent to the usual $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$ bound restricted only to excursions with $x_1 \le x_0^A$ for any fixed $A > 1$; in particular, it is enough to control the excursions with $x_1 \le x_0^2$. More strongly, the same reduction holds for excursions with $x_1 \le \lambda x_0$ whenever $\lambda > \lambda_* := \exp(\pi/\gamma_1) = 1.248896\ldots$. The square range is nevertheless structurally distinguished by the exact bilinear Heath–Brown identity, which expresses all $M(t)$, $x_0 < t \le x_0^2$, as a single bilinear form in the prefix $\mu(1), \ldots, \mu(x_0)$. Earlier geometric and correlation identities are corrected and sharpened, including the optimal fixed-length constant $1/12$ for the quadratic excursion energy.

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Remark 0.1. This notebook is exploratory, but all statements labelled unconditional are intended as proved statements. Version 7 uses one convention throughout: an excursion starts at the last

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zero immediately before a non-zero step, has one strict sign in its interior, and ends at the first return to zero; consecutive excursions may be separated by an arbitrary run of zeros. The lacunary counting lemmas are elementary. The arithmetic input in the localized RH criteria is Corollary 4 of Kaczorowski–Pintz [7], checked here against the primary source. Heuristics and open questions remain explicitly marked.

1. Introduction and conventions

The main object is the Mertens function M, decomposed into genuine excursions separated by zero runs, together with three area curves built from a regularized skeleton. Sections 3–7 collect exact formulas, sharp bounds and criteria for RH. Section 8 contains the two new lacunarity bounds and the resulting localized criteria for RH. Sections 9–11 retain the sieve-depth and large-prime decompositions of version 6, now interpreted in light of the higher-order Heath–Brown identities.

Convention on RH. Every statement carries one of the labels *[unconditional]*, *[under RH]* or *[heuristic]*. An unconditional equivalence with RH does not assume RH.

Excursion variables. The preferred endpoint notation in the new results is $x_{0,\nu}$ for the beginning and $x_{1,\nu}$ for the first return to zero. To preserve the notation of version 6, we also write $u_\nu = x_{0,\nu}$ and $v_\nu = x_{1,\nu}$.

Notation changes with respect to version 5. The symbol $M(x)$ denotes the classical Mertens function. The unregularized skeleton is M_{sk}, the squarefree counting function is $\mathcal{S}$, the Dirichlet energy is $\mathcal{E}$, and Θ is reserved for the supremum of the real parts of non-trivial zeta zeros when used in the oscillation section.

2. Excursions and the regularized skeleton

Remark 2.1 (Status of the excursion geometry). The segmentation of M by its zeros and the record of its extrema are not new: Hurst [5] computed $M(x)$ for all $x \le 10^{16}$, recording all zeros and all extrema. The area $\beta = \sum S_t$ and the quadratic functional $Q = \sum S_t^2$ are classical random-walk excursion functionals; see Takács [10], Nguyên Thê [8], Richard [9], Janson [6], Denisov–Kolb–Wachtel [2] and Carmona–Pétrélis [1]. What is specific to this notebook is the regularized triangular skeleton and the deterministic Möbius excursion analysis. The convention below absorbs all intermediate consecutive zeros into zero runs rather than treating them as zero-run pieces.

Let μ be the Möbius function and

$$M(x) = \sum_{n \le x} \mu(n), \qquad x \ge 0, \tag{2.1}$$

with $M(0) = 0$.

Definition 2.2 (Excursion and zero run)**.** An *excursion* is an interval

$$I_\nu = (x_{0,\nu}, x_{1,\nu}] = (u_\nu, v_\nu]$$

such that

$$M(x_{0,\nu}) = 0, \qquad M(x_{0,\nu}+1) = \sigma_\nu \in \{\pm 1\},$$
$$\sigma_\nu M(n) \ge 1 \quad (x_{0,\nu} < n < x_{1,\nu}), \qquad M(x_{1,\nu}) = 0,$$

and $x_{1,\nu}$ is the first zero after $x_{0,\nu}$. Its length is $\ell_\nu = x_{1,\nu} - x_{0,\nu} \ge 2$. Consecutive excursions may be separated by a *zero run*

$$[x_{1,\nu}, x_{0,\nu+1}],$$

a maximal interval of consecutive integers on which $M = 0$; it may consist of a single point. Thus the objects called "zero-run pieces" in version 6 are not excursions in version 7.

Inside I_ν put

$$H_\nu = \max_{u_\nu < n \le v_\nu} |M(n)|, \qquad \tau_\nu = \min\{n \in I_\nu : |M(n)| = H_\nu\}, \tag{2.2}$$

and put $a_\nu = \tau_\nu - u_\nu$, $b_\nu = v_\nu - \tau_\nu$, so that $\ell_\nu = a_\nu + b_\nu$.

Lemma 2.3 (Height and step count)**.** [unconditional] *Every excursion satisfies*

$$H_\nu \le \min(a_\nu, b_\nu) \le \frac{\ell_\nu}{2}. \tag{2.3}$$

Proof. The steps of M are $0, \pm 1$, so at least H_ν steps are needed to reach height H_ν from zero and at least H_ν more to return to zero. □

Definition 2.4 (Crossing and touching zero runs)**.** Let $R = [p, q]$ be a zero run between an excursion of sign σ_- ending at p and an excursion of sign σ_+ starting at q. It is a *crossing run* if $\sigma_-\sigma_+ = -1$ and a *touching run* if $\sigma_-\sigma_+ = +1$.

Definition 2.5 (Regularized skeleton)**.** On an excursion I_ν define

$$M_{\mathrm{s}}(t) = \begin{cases} \sigma_\nu \sqrt{H_\nu}\, \dfrac{t - u_\nu}{a_\nu}, & u_\nu \le t \le \tau_\nu, \\[2mm] \sigma_\nu \sqrt{H_\nu}\, \dfrac{v_\nu - t}{b_\nu}, & \tau_\nu \le t \le v_\nu, \end{cases} \tag{2.4}$$

and set $M_{\mathrm{s}} = 0$ on every zero run. The unregularized skeleton M_{sk} is obtained by replacing $\sqrt{H_\nu}$ by H_ν.

Definition 2.6 (Terminal block of a truncated computation)**.** Let $N \ge 2$ and let $u = \max\{n \le N : M(n) = 0\}$. If $u = N$ there is no terminal block. Otherwise M has constant sign $\sigma_{\mathrm{term}} = \operatorname{sgn} M(N)$ on $(u, N]$; put

$$H_{\mathrm{term}} = \max_{u < n \le N} |M(n)|, \qquad \tau_{\mathrm{term}} = \min\{n \in (u, N] : |M(n)| = H_{\mathrm{term}}\}.$$

On $[u, N]$ set

$$M_{\mathrm{s}}(x) = \begin{cases} \sigma_{\mathrm{term}} \sqrt{H_{\mathrm{term}}}\, \dfrac{x - u}{\tau_{\mathrm{term}} - u}, & u \le x \le \tau_{\mathrm{term}}, \\[2mm] \sigma_{\mathrm{term}} \left[\sqrt{H_{\mathrm{term}}} + (\sqrt{|M(N)|} - \sqrt{H_{\mathrm{term}}})\, \dfrac{x - \tau_{\mathrm{term}}}{N - \tau_{\mathrm{term}}} \right], & \tau_{\mathrm{term}} \le x \le N, \end{cases}$$

where the second line is omitted when $\tau_{\mathrm{term}} = N$. Its signed area is

$$\Delta\mathcal{A}(u, N] = \frac{\sigma_{\mathrm{term}}}{2} \left[\sqrt{H_{\mathrm{term}}}(N - u) + \sqrt{|M(N)|}(N - \tau_{\mathrm{term}}) \right].$$

This terminal skeleton is provisional: it need not coincide with the restriction of the complete skeleton after the next return to zero.

Definition 2.7 (The three curves)**.** With $M_{\mathrm{s}}^{\pm} = \max(\pm M_{\mathrm{s}}, 0)$,

$$\mathcal{A}_+(x) = \int_0^x M_{\mathrm{s}}^{+}(t)\, dt, \qquad \mathcal{A}_-(x) = -\int_0^x M_{\mathrm{s}}^{-}(t)\, dt, \qquad \mathcal{A}(x) = \int_0^x M_{\mathrm{s}}(t)\, dt. \tag{2.5}$$

The blue curve $\mathcal{A}_+$ is nondecreasing, the red curve $\mathcal{A}_-$ is nonincreasing, and all three are constant on zero runs.

Remark 2.8 (The family of g-skeletons). For any $g : \mathbb{N}_0 \to [0, \infty)$ with $g(0) = 0$ and $g(h) > 0$ for $h \ge 1$, one may replace $\sqrt{H_\nu}$ by $g(H_\nu)$. Then the signed area of a complete excursion is $\frac{1}{2}\sigma_\nu g(H_\nu)\ell_\nu$, the critical-point geometry below is unchanged, and the quadratic statements of Section 4 do not involve g at all.

3. Exact formulas

Status. Original observations with complete proofs. Lemma 3.3 corrects the corresponding lemma of version 5.

Lemma 3.1 (Increment per excursion)**.** [unconditional] *For every complete excursion,*

$$\mathcal{A}(v_\nu) - \mathcal{A}(u_\nu) = \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}. \tag{3.1}$$

The discrete version agrees:

$$\sum_{n\in I_\nu} M_{\mathrm{s}}(n) = \sigma_\nu\sqrt{H_\nu}\,\frac{(a_\nu+1)+(b_\nu-1)}{2} = \sigma_\nu\sqrt{H_\nu}\,\frac{\ell_\nu}{2}. \tag{3.2}$$

Proof. For zero-run pieces both sides vanish. Otherwise the two pieces of M_{s} are triangles of height $\sqrt{H_\nu}$ with bases a_ν and b_ν. In the discrete sum the ascent contributes $\sqrt{H_\nu}\sum_{j=1}^{a_\nu} j/a_\nu = \sqrt{H_\nu}(a_\nu+1)/2$ and the descent $\sqrt{H_\nu}\sum_{j=0}^{b_\nu-1} j/b_\nu = \sqrt{H_\nu}(b_\nu-1)/2$. □

Proposition 3.2 (Explicit form of the three curves)**.** [unconditional] *Let $x \in I_\nu$ with I_ν non-null. Then*

$$\mathcal{A}(x) = \begin{cases} \mathcal{A}(u_\nu) + \sigma_\nu\sqrt{H_\nu}\dfrac{(x-u_\nu)^2}{2a_\nu}, & u_\nu \le x \le \tau_\nu, \\[2ex] \mathcal{A}(u_\nu) + \sigma_\nu\sqrt{H_\nu}\left(\dfrac{a_\nu}{2} + (x-\tau_\nu) - \dfrac{(x-\tau_\nu)^2}{2b_\nu}\right), & \tau_\nu \le x \le v_\nu, \end{cases} \tag{3.3}$$

and, at the zeros of M,

$$\mathcal{A}(v_k) = \sum_{\nu\le k} \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \tag{3.4}$$

$$\mathcal{A}_+(v_k) = \sum_{\substack{\nu\le k \\ \sigma_\nu=+1}} \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \tag{3.5}$$

$$\mathcal{A}_-(v_k) = -\sum_{\substack{\nu\le k \\ \sigma_\nu=-1}} \frac{\sqrt{H_\nu}\,\ell_\nu}{2}. \tag{3.6}$$

The three curves are piecewise quadratic, of class C^1, and determined by the data $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$.

Proof. Integrate (2.4); for the descent use $v_\nu - t = b_\nu - (t-\tau_\nu)$. The formulas at the zeros follow from Lemma 3.1. The derivative $\mathcal{A}' = M_{\mathrm{s}}$ is continuous. □

Lemma 3.3 (Critical geometry of $\mathcal{A}$)**.** *Let $R = [p, q]$ be a zero run between an excursion of sign σ_- ending at p and an excursion of sign σ_+ starting at q.*

(i) *Every apex τ_ν is a strict inflection point:* $\operatorname{sgn}\mathcal{A}''$ *jumps from σ_ν to $-\sigma_\nu$ there.*
(ii) *If R is crossing and $p = q$, $\mathcal{A}$ has a strict local extremum at p. If $p < q$, $\mathcal{A}$ is constant on R and R is a plateau of local maxima or minima; no point of the plateau is a strict local extremum.*
(iii) *If R is touching, $\mathcal{A}$ has no local extremum on R. Its second derivative has sign $-\sigma_-$ immediately to the left of p, is zero on the interior of a non-trivial zero run, and has sign $+\sigma_+$ immediately to the right of q. If $p = q$ this is a strict inflection point; if $p < q$ the change of concavity is carried by the whole plateau.*
(iv) *Thus strict local extrema occur exactly at isolated crossing zeros, and plateaus of extrema exactly at crossing zero runs of positive length.*

Proof. On the descending side of the left excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_-$ and $\operatorname{sgn}\mathcal{A}'' = -\sigma_-$. On the ascending side of the right excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_+$ and $\operatorname{sgn}\mathcal{A}'' = \sigma_+$. On the zero run, $\mathcal{A}' = 0$. The four assertions follow by distinguishing $\sigma_+ = -\sigma_-$ from $\sigma_+ = \sigma_-$ and whether $p = q$. □

Proof. (i) and (ii) follow from $\mathcal{A}' = M_{\mathrm{s}}$ and the sign of M_{s} on each excursion. For (iii), on the ascent of I_ν one has $\mathcal{A}'' = \sigma_\nu\sqrt{H_\nu}/a_\nu$ and on the descent $\mathcal{A}'' = -\sigma_\nu\sqrt{H_\nu}/b_\nu$, which have opposite signs. Across a run of zeros $\mathcal{A}''$ passes from $-\sigma_{\nu_-}\sqrt{H_{\nu_-}}/b_{\nu_-}$ to $\sigma_{\nu_+}\sqrt{H_{\nu_+}}/a_{\nu_+}$ (with value 0 on intermediate zero-run pieces); the sign changes if and only if $\sigma_{\nu_+} = \sigma_{\nu_-}$. □

Remark 3.4 (Status of the version 5 statement). Version 5 stated that a touching zero is an inflection point of $\mathcal{A}$. This is literally true for an isolated touching zero. For a touching run of positive length, $\mathcal{A}''$ has signs $-\sigma, 0, +\sigma$ on three consecutive pieces; no single point of the plateau has strict opposite concavity on both sides.

4. Internal correlations of an excursion

Status. The triangular identity is an elementary identity for arbitrary real sequences. The excursion consequences and the sharp (L, H) bounds below are proved exactly. The optimal fixed-length constant in the upper bound for Q is $1/12$.

Let $L \geq 1$, let $\xi_1, \ldots, \xi_L \in \mathbb{R}$, put $S_0 = 0$, $S_t = \sum_{j\leq t}\xi_j$, and

$$C_h = \sum_{j=1}^{L-h} \xi_j\xi_{j+h} \qquad (1 \leq h < L).$$

Lemma 4.1 (Triangular identity)**.**

$$\sum_{t=1}^{L-1} S_t^2 + \sum_{h=1}^{L-1} hC_h = S_L \sum_{t=1}^{L-1} S_t. \tag{4.1}$$

Proof. One has $\sum_{t=1}^{L} S_t^2 = \sum_{i,j}\xi_i\xi_j(L + 1 - \max(i, j))$. Using $\max(i, j) = \frac{1}{2}(i + j + |i - j|)$ and $\sum_i i\xi_i = LS_L - \sum_{t<L} S_t$ gives $\sum_{t=1}^{L} S_t^2 = S_L^2 + S_L\sum_{t<L} S_t - \sum_h hC_h$; subtract S_L^2. □

Corollary 4.2 (Bridge case)**.** *If $S_L = 0$, then*

$$\sum_{t=1}^{L-1} S_t^2 = -\sum_{h=1}^{L-1} hC_h.$$

Remark 4.3 (Hermitian version). For complex ξ_j, define $C_h = \sum_j \xi_j\overline{\xi_{j+h}}$. Then

$$\sum_{t=1}^{L-1} |S_t|^2 + \sum_{h=1}^{L-1} h\Re C_h = \Re\left(\overline{S_L}\sum_{t=1}^{L-1} S_t\right).$$

For a complete excursion $I_\nu = (u_\nu, v_\nu]$, orient the steps by

$$\xi_j = \sigma_\nu\mu(u_\nu + j), \qquad S_t = \sum_{j=1}^{t}\xi_j = |M(u_\nu + t)|,$$

so $S_t \geq 1$ for $1 \leq t < \ell_\nu$ and $S_{\ell_\nu} = 0$. Put

$$Q_\nu = \sum_{t=1}^{\ell_\nu-1} S_t^2, \qquad C_h^{(\nu)} = \sum_{j=1}^{\ell_\nu-h} \xi_j\xi_{j+h}. \tag{4.2}$$

Then

$$Q_\nu = -\sum_{h=1}^{\ell_\nu-1} hC_h^{(\nu)}. \tag{4.3}$$

Lemma 4.4 (Zeroth moment of the internal correlations)**.** *If $q_\nu = \sum_{j=1}^{\ell_\nu} \xi_j^2$ is the number of non-zero steps, then*

$$\sum_{h=1}^{\ell_\nu - 1} C_h^{(\nu)} = -\frac{q_\nu}{2}. \tag{4.4}$$

Proof. Expand $(\sum_j \xi_j)^2 = 0$. □

Lemma 4.5 (Extreme lag)**.** *For every complete excursion,*

$$C_{\ell_\nu - 1}^{(\nu)} = -1,$$

and q_ν is even and at least 2.

Proof. The oriented first step is $\xi_1 = 1$. Immediately before the first return to zero the oriented height is 1, so the final step is $\xi_{\ell_\nu} = -1$. Hence their product is -1. Since the total step sum is zero, the numbers of $+1$ and -1 steps are equal. □

Proposition 4.6 (Quadratic decomposition by excursions)**.** *For real $x \geq 1$, let $u(x) = \max\{n \leq \lfloor x \rfloor : M(n) = 0\}$ and $L_x = \lfloor x \rfloor - u(x)$. For the terminal block put $\xi_j = \mu(u(x)+j)$, $S_t = M(u(x)+t)$ and $C_h^{\mathrm{term}} = \sum_{j=1}^{L_x - h} \xi_j \xi_{j+h}$. Then*

$$\sum_{n \leq x} M(n)^2 = \sum_{\nu:\, v_\nu \leq u(x)} Q_\nu + T(x),$$

where

$$T(x) = \sum_{t=1}^{L_x} S_t^2 = M(\lfloor x \rfloor)^2 + M(\lfloor x \rfloor) \sum_{t=1}^{L_x - 1} M(u(x)+t) - \sum_{h=1}^{L_x - 1} h C_h^{\mathrm{term}},$$

and $T(x) = 0$ if $L_x = 0$.

Proof. Every integer before $u(x)$ is either in a zero run, where its contribution is zero, or in the interior of a unique complete excursion, where Corollary 4.2 applies. The remaining block is evaluated by Lemma 4.1 with its non-zero terminal sum retained. □

Remark 4.7 (Blindness to pairs crossing excursion boundaries). For integer x,

$$\sum_{n \leq x} M(n)^2 = \sum_{a,b \leq x} \mu(a)\mu(b)(x + 1 - \max(a,b)).$$

If a lies in a completed excursion strictly before the block containing b, the weight is independent of a and $\sum_{a \in I_\nu} \mu(a) = 0$; hence cross-excursion pairs cancel exactly.

Let

$$\beta_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t = \sum_{u_\nu < n < v_\nu} |M(n)|. \tag{4.5}$$

Proposition 4.8 (Sharp bounds at fixed length and height)**.** *Let $S_0 = S_L = 0$, $S_t \geq 1$ for $0 < t < L$, $S_t - S_{t-1} \in \{-1, 0, 1\}$ and $H = \max S_t$. Then $L \geq 2H$ and*

$$H^2 + L - 2H \leq \beta \leq H(L - H), \tag{4.6}$$

$$\frac{2H^3 + H}{3} + L - 2H \leq Q \leq \frac{2H^3 + H}{3} + (L - 2H)H^2. \tag{4.7}$$

All four bounds are sharp for every $L \geq 2H \geq 2$. In particular $Q \geq (2H^3 + H)/3$ and $Q \geq L - 1$.

Proof. Before and after the first visit to height H, the path must visit each level $1, \ldots, H-1$. These $2H - 1$ forced interior times contribute H^2 to β and $(2H^3 + H)/3$ to Q. The remaining $L - 2H$ interior times lie between 1 and H, which gives the four bounds. The lower bounds are attained by $(+1, 0^{L-2H}, +1^{H-1}, -1^H)$ and the upper bounds by $(+1^H, 0^{L-2H}, -1^H)$. □

Corollary 4.9 (Fixed length)**.** *For $L \geq 2$,*

$$\beta \leq \left\lfloor \frac{L^2}{4} \right\rfloor, \qquad Q \leq \begin{cases} \dfrac{L(L^2+2)}{12}, & L \text{ even}, \\ \dfrac{L(L^2-1)}{12}, & L \text{ odd}, \end{cases}$$

and both bounds are attained by the tent profile $S_t = \min(t, L-t)$ (with one zero step at the top when L is odd). In particular $Q \leq (L^3+2L)/12$.

Corollary 4.10 (Cauchy–Schwarz, sharp form)**.**

$$\beta_\nu^2 \leq (\ell_\nu - 1) Q_\nu,$$

with equality exactly when the interior profile is constant, necessarily at height 1*.*

To separate support from sign geometry, let $P_h^{(\nu)} = \#\{j : \xi_j \xi_{j+h} \neq 0\}$ and let $D_h^{(\nu)}$ count those pairs with opposite signs. Then

$$C_h^{(\nu)} = P_h^{(\nu)} - 2D_h^{(\nu)}, \tag{4.8}$$

and

$$Q_\nu = \sum_{h=1}^{\ell_\nu - 1} h(2D_h^{(\nu)} - P_h^{(\nu)}). \tag{4.9}$$

5. Where ζ enters

Status. (5.1) is standard [16, §14.27]. (5.3) is obtained formally by the same contour method; its residues are checked, the contour details are not written out. Proposition 5.1 is original, with proof.

The black curve depends on M nonlinearly, through $\sqrt{H_\nu}$ and through the selection of u_ν, τ_ν, v_ν. No closed expression of its Mellin transform in terms of ζ is known to the author (version 5 asserted that none exists; that assertion is unproved and is withdrawn). The nodes u_ν, v_ν are zeros of M, and *[under RH and simple zeros]*

$$M(x) = \sum_\rho \frac{x^\rho}{\rho\, \zeta'(\rho)} - 2 + \sum_{n \geq 1} \frac{(-1)^{n-1} (2\pi/x)^{2n}}{(2n)!\, n\, \zeta(2n+1)}, \tag{5.1}$$

for non-integer x, the sum over ρ taken along a suitable sequence $|\gamma| < T_k \to \infty$ [16, §14.27]. For $\widetilde{A}(x) := \int_0^x M(t)\, dt$ one has, for $\operatorname{Re} s > 1$,

$$\int_1^\infty \widetilde{A}(x) x^{-s-2}\, dx = \frac{1}{s(s+1)\zeta(s)}. \tag{5.2}$$

Shifting the contour as in [16, §14.27] and collecting the residues at ρ, at $s = 0$ ($\zeta(0) = -\frac{1}{2}$) and at $s = -1$ ($\zeta(-1) = -\frac{1}{12}$), one formally arrives at *[under RH and simple zeros]*

$$\widetilde{A}(x) = \sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)\zeta'(\rho)} - 2x + 12 + O(x^{-1}). \tag{5.3}$$

The factor $(\rho+1)^{-1}$ improves convergence subject to the size of $1/\zeta'(\rho)$; hypotheses of Gonek–Hejhal type controlling this are discussed in [14]. The area of M_{sk} differs from $\widetilde{A}$ by explicit local terms:

Proposition 5.1 (Dipole identity)**.** [unconditional] *For each excursion let its* legs *be the integer sets* $\lambda = (u_\nu, \tau_\nu] \cap \mathbb{Z}$ *and* $(\tau_\nu, v_\nu] \cap \mathbb{Z}$*, with midpoints* $c_\lambda = (u_\nu + \tau_\nu)/2$ *and* $(\tau_\nu + v_\nu)/2$*. Then for every zero* v_k*,*

$$\int_0^{v_k} M_{\mathrm{sk}}(t)\,dt = \widetilde{A}(v_k) + \sum_{\lambda \subset (0, v_k]} \sum_{m \in \lambda} \mu(m)\,(m - c_\lambda). \tag{5.4}$$

Proof. On an ascent leg, $\int_u^\tau M_{\mathrm{sk}} = \sigma H a/2$ and $\int_u^\tau M = \sum_{m=u+1}^{\tau} \mu(m)(\tau - m)$, while $\sigma H = \sum_{m=u+1}^{\tau} \mu(m)$; the difference is $\sum_m \mu(m)(a/2 - \tau + m) = \sum_m \mu(m)(m - c_\lambda)$. On a descent leg, $\int_\tau^v M_{\mathrm{sk}} = \sigma H b/2$ and $\int_\tau^v M = b\,\sigma H + \sum_{m=\tau+1}^{v} \mu(m)(v - m)$ with $\sum_{m=\tau+1}^{v} \mu(m) = -\sigma H$; the difference is $\sum_m \mu(m)(m - (v - b/2))$. Sum over legs. □

6. Bounds

Status. Original, elementary; the input bounds for M are standard [17, 15, 12]. Lemma 6.1 corrects the hypothesis of version 5; Remarks 6.4 and 6.5 correct overstatements.

Lemma 6.1 (Transfer)**.** [unconditional] *Let* $x \geq 1$ *and let* $x^* \geq x$ *be the first zero of* M *not smaller than* x *(or* N *in the terminal block). If* $|M(n)| \leq c\,n^{1/2}$ *for all* $1 \leq n \leq x^*$ *(so necessarily* $c \geq 1$*), then*

$$|M_{\mathrm{s}}(t)| \leq c^{1/2} t^{1/4} \quad (0 \leq t \leq x), \qquad \mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq \tfrac{4}{5}\, c^{1/2} x^{5/4}. \tag{6.1}$$

Proof. For $t \geq \tau_\nu$, $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq c^{1/2}\tau_\nu^{1/4} \leq c^{1/2} t^{1/4}$. For $u_\nu \leq t \leq \tau_\nu$ the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$ lies below the concave function $c^{1/2}t^{1/4}$, which is ≥ 0 at u_ν and $\geq \sqrt{H_\nu}$ at $\tau_\nu \leq x^*$. Integrate. □

Corollary 6.2. [unconditional] *From the Vinogradov–Korobov bound* [17]

$$M(x) \ll x \exp\!\big(-c(\log x)^{3/5}(\log\log x)^{-1/5}\big) \tag{6.2}$$

one obtains

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{3/2} \exp\!\big(-\tfrac{c}{2}(\log x)^{3/5}(\log\log x)^{-1/5}\big). \tag{6.3}$$

Corollary 6.3. [unconditional implication] *If RH holds, then*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{5/4+\varepsilon}. \tag{6.4}$$

Consequently, if $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \gg x^{5/4+\varepsilon}$ *along a sequence, RH is false.*

Remark 6.4 (One direction only, as far as known). Since $\ell_\nu \geq 2H_\nu$, a bound $\sqrt{H_\nu}\ell_\nu \ll x^{5/4+\varepsilon}$ yields only $H_\nu \ll x^{5/6+\varepsilon}$, hence only the zero-free half-plane $\operatorname{Re} s > 5/6$. Version 5 called the black curve "a refutation test, not a criterion". Whether (6.4) is equivalent to RH is *unresolved*; the direct route fails, which is all that is shown.

Remark 6.5 (Oscillation of the heights). Since $\limsup M(x)x^{-1/2} > 1.06$ and $\liminf M(x)x^{-1/2} < -1.009$ [15] (improved in [12]), there are infinitely many n with $M(n) > 1.06\sqrt{n}$; the excursion containing n has $u_\nu < n$, so $H_\nu > 1.06\,u_\nu^{1/2}$ for infinitely many positive excursions, and similarly for negative ones. Version 5 claimed $H_\nu \gg v_\nu^{1/2-\varepsilon}$; this is *not justified*, since v_ν may be much larger than n.

7. Criteria equivalent to RH

Status. The Mellin identity (7.1) is standard [13]; the equivalences, pruning results and Corollary 7.10 are original, with proofs. Heuristics 7.5 and 7.9 are heuristic.

For $\operatorname{Re} s > 1$,

$$\int_1^\infty M(x)x^{-s-1}\,dx = \frac{1}{s\,\zeta(s)}. \tag{7.1}$$

If $\int_1^\infty |M(x)|x^{-\sigma-1}dx < \infty$ for every $\sigma > 1/2$, the left side continues analytically to $\operatorname{Re} s > 1/2$, so ζ has no zeros there.

Proposition 7.1 (L^1 criterion)**.** [unconditional] *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{v_\nu \le x} \beta_\nu + \beta_{\mathrm{term}}(x) = \sum_{n\le x} |M(n)| \ll x^{3/2+\varepsilon}. \tag{7.2}$$

Proof. RH gives $M(x) \ll x^{1/2+\varepsilon}$ [16, §14.25]. Conversely, by partial summation (7.2) gives $\sum_n |M(n)|n^{-\sigma-1} < \infty$ for $\sigma > 1/2+\varepsilon$, and M is constant on $[n, n+1)$; apply the remark after (7.1). □

Lemma 7.2 (Pruning)**.** [unconditional] *$\beta_\nu \le H_\nu \ell_\nu \le \ell_\nu^2/2$, hence*

$$\sum_{v_\nu \le x,\ \ell_\nu \le L_0} \beta_\nu \le \frac{xL_0}{2}. \tag{7.3}$$

Corollary 7.3 (Reduction to long excursions)**.** [unconditional] *For fixed $\delta \ge 0$ and $L_0 = x^{1/2-\delta}$, RH holds if and only if*

$$\sum_{v_\nu \le x,\ \ell_\nu > L_0} \beta_\nu + \beta_{\mathrm{term}}(x) \ll x^{3/2+\varepsilon} \quad \text{for every } \varepsilon > 0, \tag{7.4}$$

where the terminal block is included when its length exceeds L_0. At most x/L_0 excursions are involved.

Remark 7.4 (The $\sqrt{x}$ frontier). The cut may even depend on ε: with $L_0 = x^{1/2+\varepsilon/2}$ the discarded part is $\le x^{3/2+\varepsilon/2}/2$. The frontier is therefore $x^{1/2+o(1)}$; what requires new information is a cut x^θ with a fixed $\theta > 1/2$ (version 5 stated the frontier as $x^{1/2}$).

Heuristic 7.5 (Renewal tail)**.** This is a heuristic argument. If excursion lengths behave like those of a recurrent random walk,

$$\#\{v_\nu \le x :\ \ell_\nu > L\} \asymp \sqrt{x}\,L^{-1/2}, \tag{7.5}$$

and with $\beta_\nu \asymp c_\beta \ell_\nu^{3/2}$ the dyadic range $[L, 2L]$ contributes $\asymp \sqrt{x}\,L$ to (7.2), so the terminal scales dominate.

Remark 7.6 (Limit of pruning). For $\ell_\nu \asymp x$ the trivial bound gives $\beta_\nu \ll x^2$ against a budget $x^{3/2+\varepsilon}$: a factor $x^{1/2}$ is missing, and there are only $O(1)$ such excursions.

Theorem 7.7 (Correlation criterion)**.** [unconditional] *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{n\le x} M(n)^2 = \sum_{v_\nu \le x} Q_\nu + Q_{\mathrm{term}}(x) = -\sum_{v_\nu \le x}\sum_h h C_h^{(\nu)} + Q_{\mathrm{term}}(x) \ll x^{2+\varepsilon}. \tag{7.6}$$

Proof. RH gives $M \ll x^{1/2+\varepsilon}$. Conversely Cauchy–Schwarz gives $\sum_{n\le x}|M(n)| \le x^{1/2}(\sum M^2)^{1/2} \ll x^{3/2+\varepsilon/2}$; apply Proposition 7.1. □

Corollary 7.8 (Local sufficient condition, version 5)**.** [unconditional implication] *If for every $\varepsilon > 0$ there is C_ε with*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \textit{for every excursion (including the terminal block)}, \tag{7.7}$$

then RH holds.

Heuristic 7.9 (The hypothesis of Corollary 7.8 fails)**.** This is a heuristic argument. In a model where, after a zero of M, the next values of μ behave locally at random with squarefree density $6/\pi^2$, the pattern "k non-zero steps of one sign followed by k of the other" follows a given zero with probability $\ge c_0^{-k}$. Since M has infinitely many zeros, excursions with $H_\nu = k$ and $\ell_\nu \asymp k$ should occur with $k \to \infty$. By (**??**) $Q_\nu/\ell_\nu^{2+\varepsilon} \gg k^{1-\varepsilon} \to \infty$, so (7.7) fails for every $\varepsilon \in (0,1)$; for $\varepsilon \ge 1$ it is trivially true by Corollary 4.9. Version 5 called the condition "sufficient but not weak"; heuristically it is never satisfied.

Corollary 7.10 (Pruned local condition)**.** [unconditional implication] *If for every $\varepsilon > 0$ there is C_ε with*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \textit{whenever } \ell_\nu > v_\nu^{1/2} \tag{7.8}$$

(and likewise for the terminal block when its length exceeds $x^{1/2}$), then RH holds.

Proof. By Corollary 4.9, excursions with $\ell_\nu \le v_\nu^{1/2} \le x^{1/2}$ satisfy $Q_\nu \le (\ell_\nu^3 + 2\ell_\nu)/12 \le \ell_\nu(x+2)/12$; summing their lengths gives at most $x(x+2)/12 = O(x^2)$. The others contribute $\le C_\varepsilon \sum \ell_\nu^{2+\varepsilon} \le C_\varepsilon x^{2+\varepsilon}$. Apply Theorem 7.7. □

The ballistic excursions of Heuristic 7.9 have $\ell_\nu \asymp k \ll v_\nu^{1/2}$ and do not contradict (7.8).

Remark 7.11 (Scale of the correlation sums)*.* Under the crude model in which "j squarefree" and "$j+h$ squarefree" are independent with probability $6/\pi^2$ each, $P_h \approx (6/\pi^2)^2(L-h)$ and

$$\sum_{h<L} hP_h \approx \Big(\frac{6}{\pi^2}\Big)^2 \sum_{h<L} h(L-h) \approx \frac{6}{\pi^4} L^3 \approx 0.0616\, L^3. \tag{7.9}$$

Version 5 gave $\frac{1}{2}(6/\pi^2)^2 L^3 \approx 0.18 L^3$, omitting the factor $L-h$; that constant is *incorrect*, the order L^3 stands. For $|\sum_h hC_h|$ of size $B(\sum_h h^2)^{1/2} \asymp BL^{3/2}$ to be $O(L^2)$ one needs $B = O(\sqrt{L})$ (a calibration, not a proof).

Remark 7.12 (Conditioning on the bridge)*.* Inside an excursion the correlations are constrained by (4.3), (4.4) and by the geometry of the bridge; with the defect $d_\nu := \ell_\nu/2 - H_\nu \ge 0$, a third constraint of this type is a research direction, not a result.

Theorem 7.13 (Energy criterion)**.** [unconditional] *Define, with u the last zero $\le x$ and $H_{\mathrm{run}}(x) = \max_{u<n\le x} |M(n)|$,*

$$\mathcal{E}(x) := \frac{1}{4}\Big(\sum_{v_\nu \le x} H_\nu \ell_\nu + (x-u) H_{\mathrm{run}}(x) \Big). \tag{7.10}$$

For a complete excursion $\frac{1}{4} H_\nu \ell_\nu = \int_{I_\nu} (J')^2$, where J interpolates $\mathcal{A}$ linearly between consecutive zeros; for the terminal block this interpretation does not hold and (7.10) *is a definition. Then*

$$\sum_{n\le x} |M(n)| \le 4\mathcal{E}(x) \le x \max_{n\le x} |M(n)|, \tag{7.11}$$

and RH holds if and only if $\mathcal{E}(x) \ll x^{3/2+\varepsilon}$ for every $\varepsilon > 0$.

Proof. $\beta_\nu \le H_\nu \ell_\nu$ and $\sum \ell_\nu \le x$; apply Proposition 7.1. □

8. Lacunary long excursions and localized criteria for RH

Status. The two counting lemmas below are elementary and use only that excursions are ordered disjoint intervals. The arithmetic input is Corollary 4 of Kaczorowski–Pintz [7], equations (2.14)–(2.15), checked against the primary source. The localized RH criteria are unconditional consequences of these two ingredients. No probabilistic model is used.

Throughout this section

$$\Theta := \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}, \qquad \gamma_1 = 14.134725\ldots,$$

where γ_1 is the smallest positive ordinate of a non-trivial zeta zero.

Definition 8.1 (Two notions of a long excursion)**.** For $A > 1$ call $I_\nu = (x_{0,\nu}, x_{1,\nu}]$ *A-power-exceptional* if $x_{1,\nu} > x_{0,\nu}^A$. For $\lambda > 1$ call it *λ-ratio-exceptional* if $x_{1,\nu} > \lambda x_{0,\nu}$. The first excursion $(0, 2]$ is ignored in these asymptotic counts.

Lemma 8.2 (Power-gap count: the $O(\log\log X)$ bound)**.** *For every fixed $A > 1$,*

$$E_A^{\mathrm{pow}}(X) := \#\{\nu : x_{0,\nu} \ge 2,\ x_{0,\nu} \le X,\ x_{1,\nu} > x_{0,\nu}^A\} \le 1 + \frac{1}{\log A}\log\frac{\log X}{\log 2}.$$

In particular the square-exceptional excursions $x_1 > x_0^2$ are $O(\log\log X)$.

Proof. If $a_1 < a_2 < \cdots$ are the relevant starting points, then zero runs can only increase the separation and $a_{j+1} \ge x_{1,j} > a_j^A$. Hence $\log a_j > A^{j-1}\log 2$. □

Lemma 8.3 (Ratio-gap count: the $O(\log X)$ bound)**.** *For every fixed $\lambda > 1$,*

$$E_\lambda^{\mathrm{rat}}(X) := \#\{\nu : x_{0,\nu} \ge 2,\ x_{0,\nu} \le X,\ x_{1,\nu} > \lambda x_{0,\nu}\} \le 1 + \frac{\log(X/2)}{\log\lambda}.$$

In particular excursions with $x_1 > 2x_0$ are at most $(\log 2)^{-1}\log X + O(1)$.

Proof. For their ordered starts $a_{j+1} \ge x_{1,j} > \lambda a_j$, so $a_j > 2\lambda^{j-1}$. □

Theorem 8.4 (Kaczorowski–Pintz large-excursion supply)**.** *For every $\eta > 0$, let $N_\eta(X)$ be the number of excursions with starting point at most X that contain a point t with*

$$|M(t)| \ge \tfrac{1}{2}t^{\Theta-\eta}.$$

Then

$$\liminf_{X\to\infty} \frac{N_\eta(X)}{\log X} \ge \frac{\gamma_1}{\pi}.$$

Proof. Corollary 4 of Kaczorowski–Pintz, with their parameter $a = 0$, applies to $M_0(x) = M(x) - 1/\zeta(0) = M(x) + 2$ and gives

$$\liminf_{X\to\infty} \frac{V(M_0, X)}{\log X} \ge \frac{\gamma_1}{\pi}$$

with combined oscillation of size $x^{\Theta-\eta}$. Thus there are $(\gamma_1/\pi - o(1))\log X$ ordered sample points at which the sign alternates and $|M(t) + 2| > t^{\Theta-\eta}$. Apart from finitely many points, $|M(t)| \ge \frac{1}{2}t^{\Theta-\eta}$ and the sign of $M(t)$ equals the sign of $M(t)+2$. Alternating signs cannot lie in the same excursion, because an excursion has one strict sign throughout its interior. Zero runs between excursions do not affect this argument. □

Theorem 8.5 (Sparse-excision criterion)**.** *Let $\mathcal{E}$ be any family of excursions satisfying*

$$E_{\mathcal{E}}(X) := \#\{\nu \in \mathcal{E} : x_{0,\nu} \le X\} = o(\log X).$$

Then RH is equivalent to the following restricted bound: for every $\varepsilon > 0$ there is C_ε such that

$$|M(t)| \le C_\varepsilon t^{1/2+\varepsilon}$$

for every t belonging to an excursion not in $\mathcal{E}$. No bound at all is required on excursions in $\mathcal{E}$.

Proof. RH gives the global bound. Conversely suppose the restricted bound holds and $\Theta > 1/2$. Choose $\eta, \varepsilon > 0$ with $1/2 + \varepsilon < \Theta - \eta$. By Theorem 8.4 there are $(\gamma_1/\pi - o(1)) \log X$ distinct large excursions up to X, while only $o(\log X)$ can lie in $\mathcal{E}$. Hence arbitrarily large points in excursions outside $\mathcal{E}$ satisfy $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$, contradicting the restricted bound. Thus $\Theta \leq 1/2$, and the functional equation gives RH. □

Corollary 8.6 (Power-localized RH criterion)**.** *For every fixed $A > 1$, RH is equivalent to the RH-size bound restricted only to excursions*

$$\boxed{x_1 \leq x_0^A.}$$

In particular,

$$\boxed{\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \text{ on all excursions with } x_1 \leq x_0^2.}$$

The excursions with $x_1 > x_0^2$ may have arbitrary height, area and energy.

Proof. Lemma 8.2 gives $E_A^{\mathrm{pow}}(X) = O_A(\log\log X) = o(\log X)$; apply Theorem 8.5. □

Theorem 8.7 (Fixed-ratio strengthening)**.** *Put*

$$\lambda_* := \exp\left(\frac{\pi}{\gamma_1}\right) = 1.248896\ldots.$$

For every fixed $\lambda > \lambda_*$, RH is equivalent to the RH-size bound restricted only to excursions

$$\boxed{x_1 \leq \lambda x_0.}$$

In particular it is enough to control excursions with $x_1 \leq 2x_0$. More quantitatively, if RH is false, then for every fixed $\eta > 0$ the number of excursions $x_1 \leq \lambda x_0$ up to X that contain a point of size $\geq \frac{1}{2} t^{\Theta-\eta}$ is at least

$$\left(\frac{\gamma_1}{\pi} - \frac{1}{\log \lambda} - o(1)\right) \log X.$$

For $\lambda = 2$ the coefficient is $3.056527\ldots$.

Proof. By Theorem 8.4 there are at least $(\gamma_1/\pi - o(1)) \log X$ distinct large excursions. By Lemma 8.3, at most $(1/\log\lambda)\log X + O(1)$ of them can satisfy $x_1 > \lambda x_0$. The displayed coefficient is positive exactly when $\lambda > \exp(\pi/\gamma_1)$. The contradiction with a uniform RH-size bound on the remaining excursions is the same as in Theorem 8.5. □

Remark 8.8 (What the two new bounds do). The $O(\log\log X)$ and $O(\log X)$ estimates are not estimates for M; they count only geometrically long excursions. Their unconditional analytic use is through the independent Kaczorowski–Pintz lower bound of order $\log X$ for alternating large oscillations. The power-gap count is negligible compared with $\log X$ for every $A > 1$. The ratio-gap count is of the same order, so its explicit constant matters and yields the threshold λ_*. No assumption on the mass distribution of the discarded excursions is used.

8.1. **The square range as an exact bilinear future of the prefix.**

Proposition 8.9 (Bilinear square-horizon identity)**.** *Let $u \geq 1$ and $1 \leq t \leq u^2$. Then*

$$M(t) = 2M(\min(t,u)) - \sum_{d,j \leq u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor.$$

If $M(u) = 0$ and $u < t \leq u^2$, this becomes

$$\boxed{M(t) = -B_u(t)}, \qquad B_u(t) := \sum_{d,j \leq u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor, \tag{8.1}$$

and $B_u(u) = 0$.

Proof. This is the case $K = 2$ of Heath–Brown's identity; equivalently, it is (2.1)–(2.2) of Helfgott–Thompson [4]. Directly, with $\mu_u = \mu \mathbf{1}_{[1,u]}$, the convolution $(\mu - \mu_u) * (\mu - \mu_u) * 1$ is supported above u^2. Expanding and using $\mu * 1 = \delta$ gives $\mu(n) = 2\mu_u(n) - (\mu_u * \mu_u * 1)(n)$ for $n \le u^2$; summing gives the formula. □

Corollary 8.10 (Exact prediction of a square-ordinary excursion)**.** *Let an excursion start at $x_0 = u$ and suppose its first return satisfies $x_1 \le u^2$. Then the complete path, its first return x_1, sign, height, first maximum, area and quadratic energy are determined by the single bilinear function $B_u(t)$ in* (8.1)*. Conversely, if $B_u(t)$ has no zero after u and before u^2, the excursion is square-exceptional.*

Remark 8.11 (Higher-order determinacy and what is special about the square)*.* For every $k \ge 1$ and $n \le u^k$,

$$\mu(n) = \sum_{j=1}^{k} (-1)^{j-1} \binom{k}{j} (\mu_u^{*j} * 1^{*(j-1)})(n),$$

the general Heath–Brown identity. Thus "determined by the prefix" is not unique to the square range. What is unique at $k = 2$ is that the whole future block is one bilinear form with the elementary kernel $\lfloor t/(dj) \rfloor$; higher orders introduce higher divisor kernels and higher degree.

Question 8.12 (The square-range excursion constraint)**.** Let $M(u) = 0$, let $x_1 \le u^2$ be the first return, and write $M(t) = -B_u(t)$ as above. The boundary and sign conditions are

$$B_u(u) = B_u(x_1) = 0, \qquad \sigma B_u(t) < 0 \quad (u < t < x_1).$$

Can the rank-one coefficient structure $\mu(d)\mu(j)$ together with the product kernel $\lfloor t/(dj) \rfloor$ force the uniform RH-size bound on such complete excursions? By Corollary 8.6, that bound on square-ordinary excursions alone is equivalent to RH. By Theorem 8.7, an even stronger reduction is available at fixed ratio; the square remains attractive because of its exact bilinear form.

8.2. **Dirichlet mass of a complete excursion: the direction $M \to \zeta$.**

Proposition 8.13 (Excursion Dirichlet block)**.** *For a complete excursion $I = (x_0, x_1]$ define*

$$\mathcal{B}_I(s) := \sum_{x_0 < n \le x_1} \frac{\mu(n)}{n^s}.$$

Then, for every complex s,

$$\mathcal{B}_I(s) = \sum_{n=x_0+1}^{x_1-1} M(n)\big(n^{-s} - (n+1)^{-s}\big). \tag{8.2}$$

For real $\sigma > 0$, all summands have the sign of the excursion and

$$|\mathcal{B}_I(\sigma)| \ge \sigma x_1^{-\sigma-1} \beta_I \ge \sigma x_1^{-\sigma-1} H_I^2. \tag{8.3}$$

Proof. Partial summation gives (8.2); the endpoint terms vanish because $M(x_0) = M(x_1) = 0$. For $\sigma > 0$, $n^{-\sigma} - (n+1)^{-\sigma} \ge \sigma x_1^{-\sigma-1}$ for $x_0 < n < x_1$. Finally any path from 0 to height H and back with steps of size at most 1 has area at least $1 + \cdots + H + (H-1) + \cdots + 1 = H^2$. □

Remark 8.14 (Why this is not a numerical statement)*.* The localized criteria are uniform in x_0 and do not use the computed range of M as a statistical sample. A potentially dangerous excursion may begin at an astronomically large x_0; the counting, bilinear and Dirichlet-block identities above remain exact there.

9. Trial-division horizon of an excursion

Status. Lemma 9.1 is standard trial division. The square range is not unique as a matter of logical determinacy (Remark 8.11); it is distinguished here because the order-2 Heath–Brown identity is bilinear. The power-gap count is already proved in Lemma 8.2.

Lemma 9.1 (Square horizon)**.** [unconditional] *Let $u \geq 2$. Trial division by the primes $p \leq u$ alone determines $\mu(n)$ for every $n \leq u^2$.*

Proof. Remove the prime factors $p \leq u$ of n; if one occurs twice, $\mu(n) = 0$. The cofactor $q \leq n \leq u^2$ has all prime factors $> u$, so it is 1 or a single prime (two such factors would give $q > u^2$). □

Warning 9.2 (On the word "information")*.* Version 5 described primes $> u$ as information not available at the start u of an excursion. In the logical sense this is *incorrect*: the primes $\leq u$ determine, by the sieve of Eratosthenes, all primes $\leq u^2$, and by iteration all primes. What Lemma 9.1 measures is *sieve depth*. In Sections 9 to 11 the words "known", "new" and "information" are used only in this sense.

Definition 9.3 (Smooth and rough parts)**.** For $y \geq 1$ write $n = s_y(n)\, r_y(n)$, where every prime factor of $s_y(n)$ is $\leq y$ and every prime factor of $r_y(n)$ is $> y$. Let $P^+(n)$ be the largest prime factor of n, with $P^+(1) = 1$.

Lemma 9.4. [unconditional]

$$\mu(n) = \mu(s_y(n))\, \mu(r_y(n)); \tag{9.1}$$

if $n < y^k$ then $\Omega(r_y(n)) \leq k - 1$, and if $n \leq y^2$ then $r_y(n)$ is 1 or a prime.

Proposition 9.5 (Forced ternary content of a complete excursion)**.** *Let $I = (x_0, x_1]$ be a complete excursion, $L = x_1 - x_0$, and let $q = \sum_{x_0 < n \leq x_1} \mu(n)^2$. Then the numbers of $+1$ and -1 steps are exactly $q/2$ each. Moreover*

$$q = \frac{6}{\pi^2} L + O(\sqrt{x_1}),$$

so

$$N_+ = N_- = \frac{3}{\pi^2} L + O(\sqrt{x_1}), \qquad N_0 = \left(1 - \frac{6}{\pi^2}\right) L + O(\sqrt{x_1}).$$

If the excursion is square-exceptional then $L > x_1 - \sqrt{x_1}$, hence these three counts are linear in x_1. Thus square-exceptionality does not mean that the long interval is supported on only a few non-zero Möbius steps.

Proof. The equality of the $+1$ and -1 counts follows from the zero net increment over a complete excursion. The asymptotic is the standard squarefree count $\sum_{n \leq x} \mu(n)^2 = 6x/\pi^2 + O(\sqrt{x})$ applied at the two endpoints. □

Proof. The two parts are coprime. If $\Omega(r) \geq 1$ then $y^{\Omega(r)} < r \leq n$. □

Definition 9.6 (Square-horizon ordinary and exceptional excursions)**.** An excursion is *square-ordinary* if $v_\nu \leq u_\nu^2$ and *square-exceptional* if $v_\nu > u_\nu^2$. Version 6 called these "square-ordinary" and "information-expanding"; version 7 avoids that wording, because higher-order Heath–Brown identities determine arbitrarily large finite ranges from a fixed prefix. The square range remains the unique order-2 bilinear range.

Proposition 9.7 (Double-exponential separation)**.** [unconditional] *If $a_1 < a_2 < \cdots$ are the starts ≥ 2 of square-exceptional excursions, then*

$$a_{j+1} > a_j^2, \qquad a_j \geq a_1^{2^{j-1}},$$

and therefore

$$\#\{j : a_j \le X\} \le 1 + \log_2 \frac{\log X}{\log 2}. \tag{9.2}$$

This is the case $A = 2$ of Lemma 8.2.

Proof. An expanding excursion starting at a_j ends beyond a_j^2, and the next one starts at or after that end. □

Question 9.8. Are there only finitely many square-exceptional excursions? Equivalently: does M vanish in $(x, x^2]$ for every sufficiently large x? (If the zeros $u \ge x_0$ all have a zero in $(u, u^2]$, apply this to the last zero $\le x$; the converse is immediate.) No unconditional result of this form has been located; results counting sign changes of arithmetic error terms and lower bounds for the mean of $|M|$ do not give it.

Remark 9.9 (The horizon u^2 is not the multiplicative threshold of an excursion). For a self-contained excursion $\sqrt{v_\nu} \le u_\nu$. Every $n \le v_\nu$ has at most one prime factor $> \sqrt{v_\nu}$, so the threshold that governs the multiplicative structure inside I_ν is $\sqrt{v_\nu}$, which lies below u_ν. The split "known/new" at u_ν does not match it, and for a self-contained excursion the regime $t > u_\nu^2$ of Section 10 lies beyond v_ν. That section is therefore about $M(t)$, not about a single excursion; Section 11 gives the local version.

10. Large-prime decomposition at level u^2

Status. All identities are original and elementary. The asymptotic statements use only the prime number theorem with classical error term and the Vinogradov–Korobov bound [13, 17], and are proved here. Several statements of version 5 are corrected; see Warnings 10.9, 10.13, 10.18 and 10.19.

Definition 10.1 (Baseline and large-prime part)**.** For $u \ge 2$ and $t \ge 1$ put

$$B_u(t) := \sum_{\substack{n \le t \\ P^+(n) \le u^2}} \mu(n), \qquad E_u(t) := \sum_{\substack{n \le t \\ P^+(n) > u^2}} \mu(n), \qquad M(t) = B_u(t) + E_u(t). \tag{10.1}$$

Version 5 used an undefined "baseline" $B_u(t)$; (10.1) is the definition adopted here. In the window $u^2 < t < u^4$, E_u is the quantity $E_u^{(1)}$ of version 5.

Proposition 10.2 (Single-prime copy)**.** [unconditional] *For a prime p and $t < p^2$,*

$$\Delta_p(t) := \sum_{a \le t/p} \mu(ap) = -M(t/p). \tag{10.2}$$

Only $p > \sqrt{t}$ is used; the condition $p > u^2$ of version 5 plays no role.

Proof. $a \le t/p < p$, so $p \nmid a$ and $\mu(ap) = -\mu(a)$. □

Proposition 10.3 (Sign content of one prime cone)**.** [unconditional] *Let $\mathcal{S}(y) = \sum_{a \le y} \mu(a)^2$. Among ap, $a \le y < p$, the numbers of values $+1, -1, 0$ of μ are*

$$N_+ = \frac{\mathcal{S}(y) - M(y)}{2}, \qquad N_- = \frac{\mathcal{S}(y) + M(y)}{2}, \qquad N_0 = \lfloor y \rfloor - \mathcal{S}(y), \tag{10.3}$$

with limiting proportions $3/\pi^2,\ 3/\pi^2,\ 1 - 6/\pi^2$.

Proposition 10.4 (Linear window)**.** [unconditional] *If $u^2 < t < u^4$, no $n \le t$ has two prime factors $> u^2$ (counted with multiplicity), and*

$$E_u(t) = -\sum_{u^2 < p \le t} M(t/p). \tag{10.4}$$

Proposition 10.5 (Prime-count form)**.** [unconditional] *For $u^2 < t < u^4$ and $R := t/u^2$,*

$$E_u(t) = -\sum_{a\le R} \mu(a)(\pi(t/a) - \pi(u^2)), \qquad M(t) = B_u(t) - \sum_{a\le R} \mu(a)\pi(t/a) + M(R)\pi(u^2). \tag{10.5}$$

Proof. Expand (10.4) as $-\sum_p \sum_{a\le t/p} \mu(a)$ and interchange: for fixed a the prime runs over $u^2 < p \le t/a$, which is non-empty only if $a < R$. □

Remark 10.6 (How the primes above u^2 enter)*.* The only input from primes $> u^2$ is the value of π at the points t/a, $a \le R$, weighted by $\mu(a)$ at the small scale $R < u^2$. Individual positions of those primes matter only through these counts.

Definition 10.7 (Copy multiplicities)**.** For $u^2 < t < u^4$, $R = t/u^2$ and $1 \le y \le \lfloor R\rfloor$ put

$$N_y(t,u) = \#\Big\{p:\ \max(u^2, \tfrac{t}{y+1}) < p \le \tfrac{t}{y}\Big\}, \qquad E_u(t) = -\sum_{1\le y\le R} N_y(t,u) M(y). \tag{10.6}$$

Remark 10.8 (Asymptotics of the multiplicities, corrected)*.* For fixed $R > 1$ and $u \to \infty$,

$$N_y \sim \frac{Ru^2}{2\log u}\cdot\frac{1}{y(y+1)} \quad (1 \le y \le R-1), \qquad N_{\lfloor R\rfloor} \sim \frac{u^2}{2\log u}\Big(\frac{R}{\lfloor R\rfloor} - 1\Big) \quad (R \notin \mathbb{Z}), \tag{10.7}$$

and $N_R = 0$ if $R \in \mathbb{Z}$.

Warning 10.9*.* Version 5 stated the first relation of (10.7) for all $y < R$. This is *incorrect* for $y = \lfloor R\rfloor$ when $R \notin \mathbb{Z}$, since then $t/(y+1) < u^2$.

Definition 10.10 (Coherent coefficient)**.**

$$\mathcal{C}(R) := \int_1^R \frac{M(w)}{w^2}\,dw = \sum_{a\le R} \mu(a)\Big(\frac{1}{a} - \frac{1}{R}\Big) = \sum_{a\le R} \frac{\mu(a)}{a} - \frac{M(R)}{R}. \tag{10.8}$$

Proposition 10.11 (Fixed R, corrected)**.** [unconditional] *For fixed $R > 1$, as $u \to \infty$,*

$$E_u(Ru^2) = -\frac{Ru^2}{2\log u}(\mathcal{C}(R) + o(1)). \tag{10.9}$$

Proof. In (10.5) the sum is finite; for $a < R$ the prime number theorem gives $\pi(Ru^2/a) - \pi(u^2) \sim (R/a - 1)u^2/(2\log u)$, and $\sum_{a\le R} \mu(a)(R/a - 1) = R\mathcal{C}(R)$. □

Lemma 10.12 (The coefficient of version 5)**.** [unconditional] *The coefficient $A(R)$ of version 5, here*

$$\Theta(R) := \sum_{1\le y\le R} \frac{M(y)}{y(y+1)} = \sum_{n\le R} \frac{\mu(n)}{n} - \frac{M(R)}{\lfloor R\rfloor + 1}, \tag{10.10}$$

satisfies

$$\Theta(R) = \mathcal{C}(\lfloor R\rfloor + 1), \qquad \mathcal{C}(R) = \Theta(R) - M(R)\Big(\frac{1}{R} - \frac{1}{\lfloor R\rfloor + 1}\Big), \tag{10.11}$$

and

$$\mathcal{C}(R) = -\int_R^\infty \frac{M(w)}{w^2}\,dw, \qquad \Theta(R) = -\sum_{n>R} \frac{M(n)}{n(n+1)}. \tag{10.12}$$

Version 5 wrote $M(R)/(R+1)$ in (10.10)*; this is correct only for integer R.*

Proof. M is constant on $[y, y+1)$ and $\int_y^{y+1} w^{-2}dw = 1/(y(y+1))$. By (7.1) and dominated convergence as $s \downarrow 1$ (using $M(w) \ll we^{-c\sqrt{\log w}}$), $\int_1^\infty M(w)w^{-2}dw = 0$. □

Warning 10.13 (Main term of version 5). Version 5 asserted $E_u^{(1)}(Ru^2) \sim -\frac{Ru^2}{2\log u}\Theta(R)$. By (10.9) and (10.11) this is *correct only when* $M(\lfloor R\rfloor) = 0$. For $1 < R < 2$ one has $E_u = -(\pi(t) - \pi(u^2)) \sim -(R-1)\frac{u^2}{2\log u}$, whereas version 5 gives $-\frac{R}{2} \cdot \frac{u^2}{2\log u}$.

Corollary 10.14 (Decay of the coefficients)**.** [unconditional] *By* (10.12) *and* (6.2)*,*

$$\mathcal{C}(R),\ \Theta(R) \ll \exp(-c_1(\log R)^{3/5}(\log\log R)^{-1/5}). \tag{10.13}$$

Proposition 10.15 (Both parts are large)**.** [unconditional]

(i) *For fixed* $R > 1$ *with* $\mathcal{C}(R) \neq 0$*,*

$$E_u(Ru^2) \sim -\frac{Ru^2}{2\log u}\mathcal{C}(R), \qquad B_u(Ru^2) \sim +\frac{Ru^2}{2\log u}\mathcal{C}(R); \tag{10.14}$$

for $1 < R < 2$*,* $\mathcal{C}(R) = 1 - 1/R > 0$*.*

(ii) *There is* $C > 0$ *such that, as* $t \to \infty$*, uniformly for* $u^2 < t < u^4$ *with* $R = t/u^2 \geq \exp(C(\log\log t)^2)$*,*

$$E_u(t) = (1+o(1))\frac{t}{(\log t)^2}, \qquad B_u(t) = -(1+o(1))\frac{t}{(\log t)^2}. \tag{10.15}$$

In both regimes $|E_u(t)|$ *and* $|B_u(t)|$ *exceed* $t^{1/2+\varepsilon}$ *for every fixed* $\varepsilon < 1/2$ *and all large* t*.*

Proof. We use [13]: for some $c > 0$,

$$\pi(y) = \mathrm{li}(y) + O(ye^{-c\sqrt{\log y}}), \quad M(y) \ll ye^{-c\sqrt{\log y}}, \quad \widetilde{m}(y) := \sum_{a\leq y}\frac{\mu(a)}{a} \ll e^{-c\sqrt{\log y}}, \tag{10.16}$$

and $\sum_{n\geq 1}\mu(n)\log n/n = -1$ (the value at $s = 1$ of $\zeta'(s)/\zeta(s)^2$). Hence

$$\int_1^\infty \widetilde{m}(w)\frac{dw}{w} = \lim_{Y\to\infty}\Big(\widetilde{m}(Y)\log Y - \sum_{a\leq Y}\frac{\mu(a)\log a}{a}\Big) = 1, \tag{10.17}$$

with absolute convergence.

(i) Proposition 10.11, and $M(t) = o(t/\log t)$. For $1 < R < 2$, $M = 1$ on $[1, 2)$.

(ii) Since $t < u^4$, $R < \sqrt{t}$, so $\log(t/a) \geq \frac{1}{2}\log t$ for $a \leq R$. By (10.5),

$$E_u(t) = -\sum_{a\leq R}\mu(a)\,\mathrm{li}(t/a) - \sum_{a\leq R}\mu(a)(\pi(t/a) - \mathrm{li}(t/a)) + M(R)\pi(u^2). \tag{10.18}$$

The second term is $\ll t(1+\log R)e^{-c\sqrt{\log t/2}} = o(t/\log^2 t)$. The third is $\ll Re^{-c\sqrt{\log R}}\cdot u^2/\log u \ll e^{-c\sqrt{\log R}}\,t/\log t$, which is $o(t/\log^2 t)$ once $c\sqrt{C} > 1$. For the first, with $F(w) = w\,\mathrm{li}(t/w)$,

$$\sum_{a\leq R}\mu(a)\,\mathrm{li}(t/a) = \widetilde{m}(R)F(R) - \int_1^R \widetilde{m}(w)F'(w)\,dw, \qquad F'(w) = \mathrm{li}(t/w) - \frac{t/w}{\log(t/w)} = \frac{t/w}{\log^2(t/w)}\Big(1+O\Big(\frac{1}{\log t}\Big)\Big). \tag{10.19}$$

The boundary term is $\ll e^{-c\sqrt{\log R}}t/\log t = o(t/\log^2 t)$. The integral equals $\frac{t}{\log^2 t}\big(\int_1^R \widetilde{m}(w)g_t(w)\frac{dw}{w} + O(1/\log t)\big)$ with $g_t(w) = \log^2 t/\log^2(t/w) \in [1, 4]$ and $g_t \to 1$ pointwise; by dominated convergence and (10.17) it is $(1 + o(1))t/\log^2 t$. Hence $E_u(t) = (1 + o(1))t/\log^2 t$, and $B_u = M - E_u$ with $M(t) = o(t/\log^2 t)$. □

Remark 10.16 (Why $t/\log^2 t$). $\mathcal{C}(R)$ captures only the $1/\log t$ term, which tends to zero. The $1/\log^2 t$ term comes from $\zeta'(s)/\zeta(s)^2 \to -1$, that is, from the pole of ζ at $s = 1$, and it does not decay in R. The version 5 claim that the "coherent fraction tends to zero" is true for the first term and does not describe the size of E_u.

Definition 10.17 (Remaining margin, version 5)**.** For a target $T(t) = t^{1/2+\varepsilon}$ put

$$G_{u,\varepsilon}(t) = \max(0,\ T(t) - |B_u(t)|), \tag{10.20}$$

and call the large-prime part *causally responsible* for a crossing of T if $|E_u(t)| \geq G_{u,\varepsilon}(t)$ at the crossing.

Warning 10.18 (The margin is vacuous)*.* With B_u as in (10.1), Proposition 10.15 gives $|B_u(t)| > T(t)$, hence $G_{u,\varepsilon}(t) = 0$, for all large t in both regimes. The test is always passed and does not discriminate. "Causal responsibility" of one of two cancelling terms of size $t/\log^k t$ for the value of their sum is not well defined.

Warning 10.19 (Calibration remarks of version 5)*.* Version 5 required $|A(R)| \ll t^{-1/2+\varepsilon}\log t$ for the large-prime part to stay at the RH scale, and for $t = u^\alpha$ derived $|A(R)| \ll R^{-\eta(\alpha,\varepsilon)}$ with $\eta = \alpha(1/2-\varepsilon)/(\alpha-2)$ ($\eta = 3/2, 1, 3/4, 5/8$ at $\alpha = 3, 4, 6, 10$). These remarks are *superseded*: the coefficient is $\mathcal{C}$, not Θ (Warning 10.13); for $\alpha > 4$ the point t lies outside the window of Proposition 10.4, where (10.2) fails for $u^2 < p \leq \sqrt{t}$; and by Proposition 10.15, E_u alone is never at the RH scale, so only $B_u + E_u = M$ can be compared with T.

Remark 10.20 (First copy transitions)*.* For $1 < R < 2$ every copy has length 1 and $E_u(t) = -(\pi(t) - \pi(u^2))$, consistent with $\mathcal{C}(R) = 1 - 1/R$. For $2 < R < 3$ copies of length 2 appear but $M(2) = 0$ ($\mathcal{C} = 1/2$). For $3 < R < 4$ copies of length 3 enter with $M(3) = -1$, of opposite net sign.

11. Local large-prime decomposition of an excursion

Status. Proposition 11.1 and Lemma 11.3 are original, with proofs; the input on primes in short intervals is from the literature [11]; Heuristic 11.4 is heuristic.

Proposition 11.1 (Local decomposition)**.** [unconditional] *Let $I_\nu = (u, v]$ be a complete excursion and $y = \sqrt{v}$. For $u < t \leq v$,*

$$M(t) = \sum_{\substack{u<n\leq t\\ P^+(n)\leq y}} \mu(n) - \sum_{a<y} \mu(a)\,\Pi_a(t), \qquad \Pi_a(t) := \#\Big\{p:\ \max\big(y, \tfrac{u}{a}\big) < p \leq \tfrac{t}{a}\Big\}, \tag{11.1}$$

and in particular the bridge condition $M(v) = 0$ reads

$$\sum_{\substack{u<n\leq v\\ P^+(n)\leq \sqrt{v}}} \mu(n) = \sum_{a<\sqrt{v}} \mu(a)\,\Pi_a(v). \tag{11.2}$$

Proof. $M(u) = 0$. An $n \leq v = y^2$ has at most one prime factor $p > y$; then $n = ap$ with $a < v/y = y < p$, so $\mu(n) = -\mu(a)$, and $(a, p) \mapsto ap$ is a bijection onto these n. □

Remark 11.2 (What the large primes do inside an excursion)*.* The decomposition holds for every excursion, with threshold $\sqrt{v_\nu}$ and no horizon u_ν^2. The large-prime input consists of prime counts in the intervals $(u/a, t/a]$, of length at most ℓ_ν/a at height about v_ν/a. The asymptotic $\pi(x) - \pi(x-h) \sim h/\log x$ is known for $x^\theta \leq h \leq x$ with fixed $\theta > 7/12$ [11], and not for shorter intervals. Already for $a = 1$, excursions with $\ell_\nu < v_\nu^{7/12}$ have large-prime input without an unconditional asymptotic.

Lemma 11.3 (Möbius-weighted inverse logarithms)**.** [unconditional] *With C as in Proposition 10.15, uniformly for $\exp(C(\log\log x)^2) \leq z \leq \sqrt{x}$,*

$$\sum_{a\leq z} \frac{\mu(a)}{a\log(x/a)} = -\frac{1+o(1)}{(\log x)^2}. \tag{11.3}$$

Proof. Partial summation gives $\widetilde{m}(z)/\log(x/z) - \int_1^z \widetilde{m}(w)\,dw/(w\log^2(x/w))$; argue as in the proof of Proposition 10.15. □

Heuristic 11.4 (Opposite drifts)**.** This is a heuristic argument. Let $u \geq \sqrt{v}\exp(C(\log\log v)^2)$. Near $t \in I_\nu$ the counts Π_a grow for $a < t/\sqrt{v}$, at rate $1/(a\log(t/a))$ if primes near t/a have their expected local density uniformly in a (beyond what is known). Ignoring these difficulties, over $(t, t+h]$ the large-prime part changes by about $-h\sum_{a<t/\sqrt{v}}\mu(a)/(a\log(t/a)) = (1+o(1))h/\log^2 t$ by Lemma 11.3. Since $M(v) - M(u) = 0$, the $\sqrt{v}$-smooth part drifts at rate about $-1/\log^2 v$. The drift does not depend on σ_ν. For $\ell_\nu \gg (\log v_\nu)^4$ both parts exceed $\sqrt{\ell_\nu}$, and (11.2) is a balance between two opposite drifts: the local analogue of Proposition 10.15.

12. A model for the black curve

Status. Heuristic throughout.

Heuristic 12.1 (Scale and period)**.** This is a heuristic argument. Motivated by (5.1), one is led to model $M(x) \approx \sqrt{x}\,F(\log x)$ with F almost periodic, dominated at low frequency by $\gamma_1 = 14.1347\ldots$. Within an excursion of length $\asymp x$ this suggests $H_\nu \asymp \sqrt{x}$ and $\sqrt{H_\nu}\ell_\nu \asymp x^{5/4}$, so the black curve should oscillate at scale

$$|\mathcal{A}(x)| \asymp x^{5/4}, \tag{12.1}$$

with a period $2\pi/\gamma_1$ in $\log x$, that is, a factor $e^{2\pi/\gamma_1} \approx 1.56$ in x.

13. Numerical data

Status. Observations recorded in version 5; no new computation was carried out for version 6.

Experiment 13.1 (Excursion census)**.** *Object:* the excursion data $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$ and the curves $\mathcal{A}_+, \mathcal{A}_-, \mathcal{A}$. *Range:* $n \leq N = 455\,052\,511$, with $Z(N) = 87\,385$ excursions. *Expected outcome:* the bound (6.1); the prediction $Z(N)N^{-1/4} \approx 6.0 \times 10^2$ excursions longer than $\sqrt{N} \approx 2.13 \times 10^4$ from Heuristic 7.5. *Observed outcome (version 5):*

$$\mathcal{A}_+(N) = 6.3449 \times 10^9, \quad \mathcal{A}_-(N) = -7.0877 \times 10^9, \quad \mathcal{A}(N) = -7.4279 \times 10^8, \tag{13.1}$$

$$\max_\nu \frac{H_\nu}{\sqrt{\ell_\nu}} = 2.0743, \qquad \max_\nu \frac{\beta_\nu}{\ell_\nu^{3/2}} = 1.0214. \tag{13.2}$$

The count of excursions longer than $\sqrt{N}$ and the maxima restricted to $\ell_\nu > \sqrt{N}$ were not recorded. At this range $N^{0.05} \approx 2.7$, so factors x^ε are not separable from constants.

Warning 13.2 (Constant in the transfer bound). Version 5 compared $|\mathcal{A}(N)|$ with (6.1) using $c = \max_{10^3 \leq n \leq N} |M(n)|n^{-1/2}$, reporting a bound about 30 times larger than $|\mathcal{A}(N)|$. That c does not satisfy the hypothesis of Lemma 6.1, which requires all $n \leq x^*$; since $M(1) = 1$, any admissible c is ≥ 1. The comparison must either use the full maximum or split off the range below 10^3.

14. Open questions and future work

Status. The first items are analytic questions; the computed range of M is not used as evidence for asymptotic representativeness.

(1) Question 8.12: exploit the complete-excursion sign and return conditions inside the bilinear form $B_u(t)$ to obtain a uniform bound on square-ordinary excursions.
(2) Determine whether the stronger fixed-ratio reduction of Theorem 8.7 admits an identity as tractable as the bilinear square identity.
(3) Analyze cancellation between the signed Dirichlet blocks $\mathcal{B}_I(s)$ of Proposition 8.13; there is no cancellation inside one excursion for real $s > 0$.
(4) Question 9.8: whether there are finitely many square-exceptional excursions. This is not needed for the localized RH criterion.
(5) Whether (6.4) is equivalent to RH (Remark 6.4).

(6) Whether the pruned local condition (7.8) holds, and whether the bridge geometry yields another exact correlation constraint beyond (4.3), (4.4), and $C_{\ell-1} = -1$.

Notebook History

Version 5. Single section: definitions, exact formulas, internal correlations, bounds, criteria, prime horizon, copy propagation, model, census data.

Version 6. Restructured into labelled sections following the authoring rules. Corrections: zero-run pieces and touching zeros (Definition 2.4, Lemma 3.3); Remark 6.5; Remark 6.4; hypothesis of Lemma 6.1; constant in Remark 7.11; $j = 1$ in Proposition 9.7; (10.7); main term (Proposition 10.11, Warning 10.13). New: Warning 9.2, Lemma 9.4, Proposition 10.5, Proposition 10.15, Section 11, Heuristic 7.9, Corollary 7.10, proofs of Propositions 7.1 and 5.1. Archived as superseded: the calibration remarks (Warning 10.19) and the margin test (Warning 10.18).

Version 7. Merged against the 1485-line version 6 source. The excursion convention is made consistent with zero runs between genuine excursions. Section 4 gains the real/Hermitian triangular identity, $C_{\ell-1} = -1$, the exact terminal quadratic term, sharp (L, H) bounds, the fixed-length constant $1/12$, and the sharp Cauchy–Schwarz factor $L - 1$. The principal new analytic section proves the power-gap $O(\log\log X)$ and ratio-gap $O(\log X)$ counts and combines them with the verified Kaczorowski–Pintz Corollary 4. This yields the sparse-excision criterion, the reduction of RH to excursions with $x_1 \le x_0^A$ for every $A > 1$ (in particular x_0^2), and the stronger fixed-ratio reduction for every $\lambda > \exp(\pi/\gamma_1)$. The square range is then represented by the exact bilinear Heath–Brown/Helfgott–Thompson identity.

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